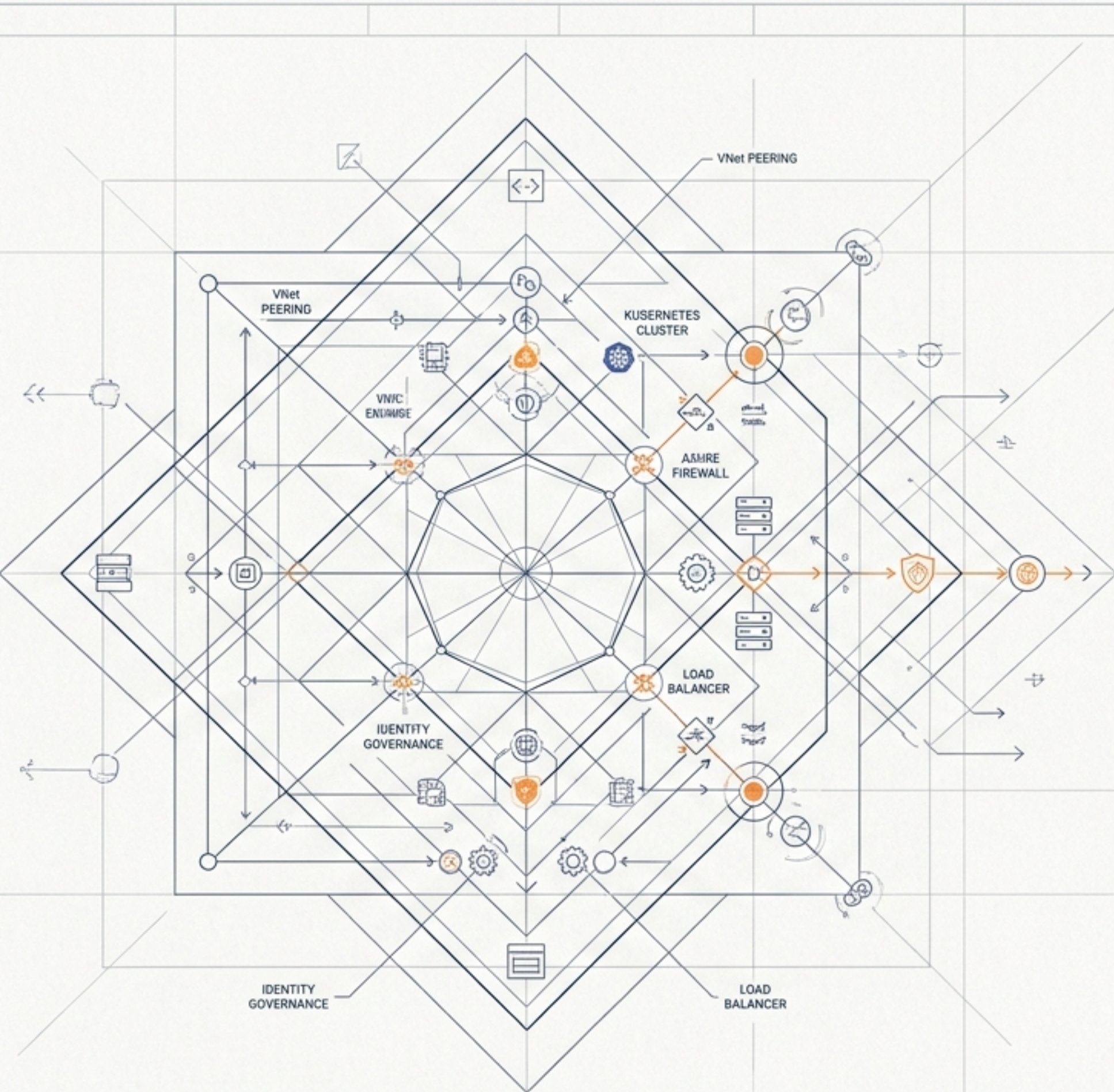


A High-Resolution Visual Guide to a Production-Ready Azure Ecosystem



# ENTERPRISE CLOUD INFRASTRUCTURE BASELINE (VISION 2026)

## ARCHITECTURAL BLUEPRINTS & DEVSECOPS PATTERNS



Compound Auth, Managed Identities, and OIDC Federation for secretless pipelines.



100% Terraform. Declarative, idempotent state with zero configuration drift.



Stateless, predictable environments  
prioritizing resource  
replacement over patching.



Secure AppLink mTLS tunnels for legacy on-premises data exchange.



Infrastructure is born with ubiquitous, attached telemetry.

| LEGEND                                                                                |       |
|---------------------------------------------------------------------------------------|-------|
|  | _____ |
|  | _____ |
|  | _____ |
|  | _____ |
|  | _____ |

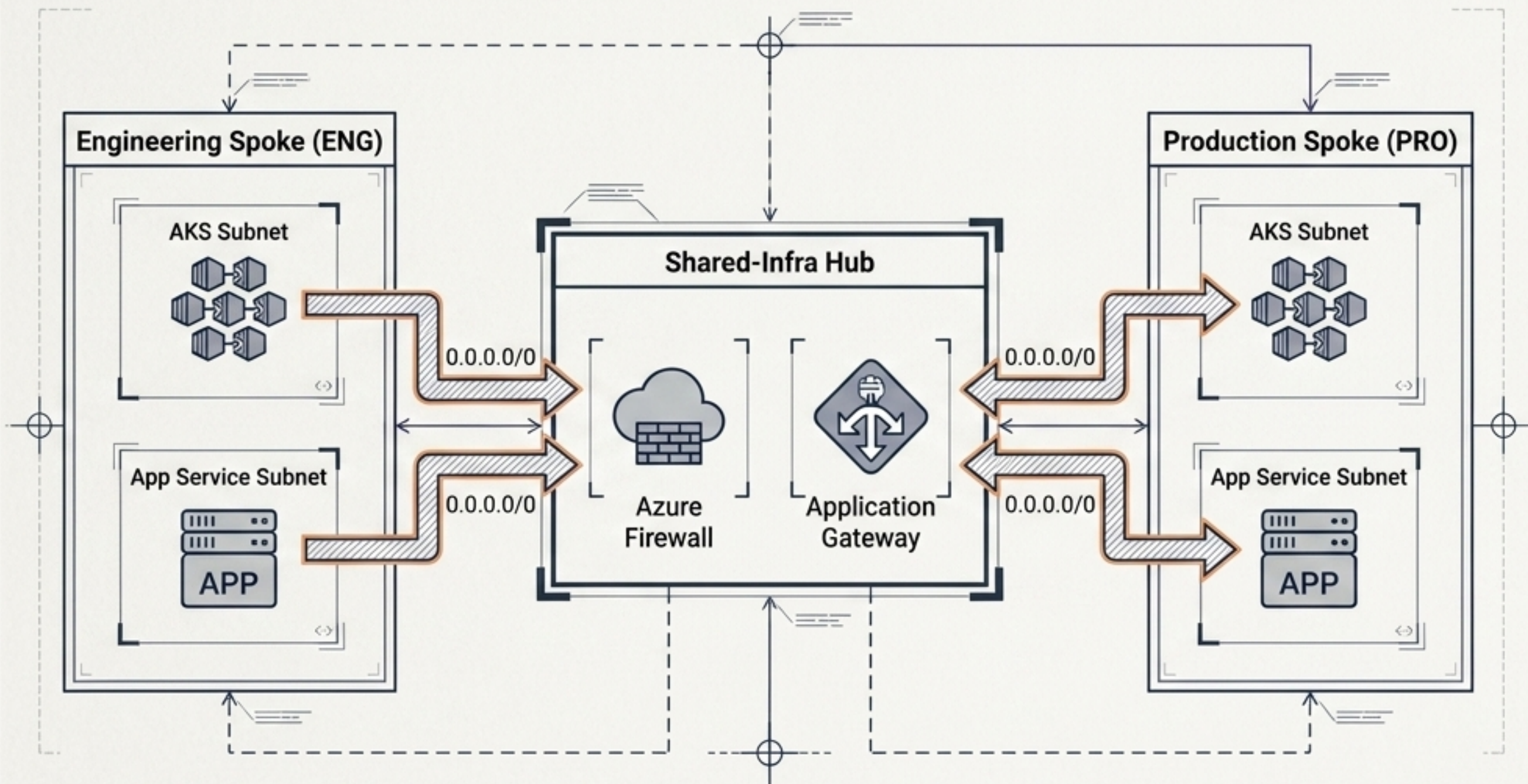
| HEERDOCTORS | ADON | SO |
|-------------|------|----|
|             |      |    |



| Dimension                    | Engineering Spoke (ENG)                                   | Production Spoke (PRO)                                |
|------------------------------|-----------------------------------------------------------|-------------------------------------------------------|
| IaC Branch & Naming Prefix   | develop branch   'd' prefix<br>(e.g., rg-appcore-dne-dev) | main branch   No prefix<br>(e.g., rg-appcore-ne-pro)  |
| App Service SKU              | B1 or P1v2                                                | Premium P2v2 or P3v3                                  |
| AKS Node Pool                | Standard_DS2_v2                                           | Standard_D4s_v3<br>(Auto-healing enabled)             |
| Persistence (MongoDB Tier)   | M10 (Standard)                                            | M30+ (High-Availability)                              |
| Storage Replication & Backup | Local Redundant (LRS)  <br>7-day retention                | Geo/Zone Redundant (GRS/ZRS)  <br>180+ days retention |



# Symmetrical Hub-Spoke Network Architecture



## Symmetrical Hub-Spoke Model

Achieves absolute environment isolation while centralizing egress security. The Hub acts as the single inspection point.

## Forced Tunneling (UDR)

Workload subnets route mapped to Azure Firewall Premium via 0.0.0.0/0.

## Edge Security

Application Gateway WAF v2 terminates SSL and delivers L7 protection over peering links.





### Multi-Region Fan-Out

Global architecture allows independent activation of geographic footprints via Terraform feature flags (e.g., `deploy_Europe = true`).

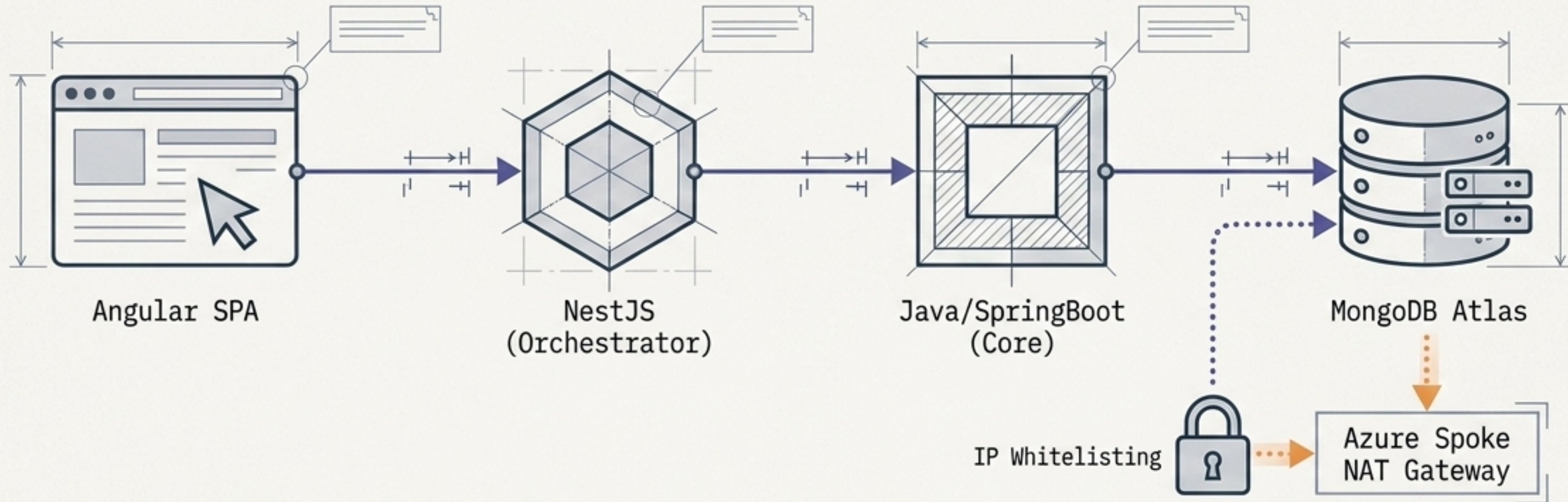
### Carbon-Aware Naming Logic

- ne (North Europe): Primary Production. Carbon Intensity: Low (Wind/Hydro).
- we (West Europe): Disaster Recovery. Carbon Intensity: Moderate.
- eus (East US): North American Edge. Carbon Intensity: High.



# Microservices & Polyglot Architecture

Viewport (70%)



Telemetry Panel (30%)

## Polyglot Architecture

- Frontend: Angular SPA
- Orchestration: NestJS (Node.js)
- Core Backend: Java/SpringBoot

## Secure MongoDB Atlas Integration

- Each Azure Spoke maps to a dedicated Atlas Project. Strict IP Whitelisting enforces traffic exclusively from corresponding Azure Spoke NAT Gateways.

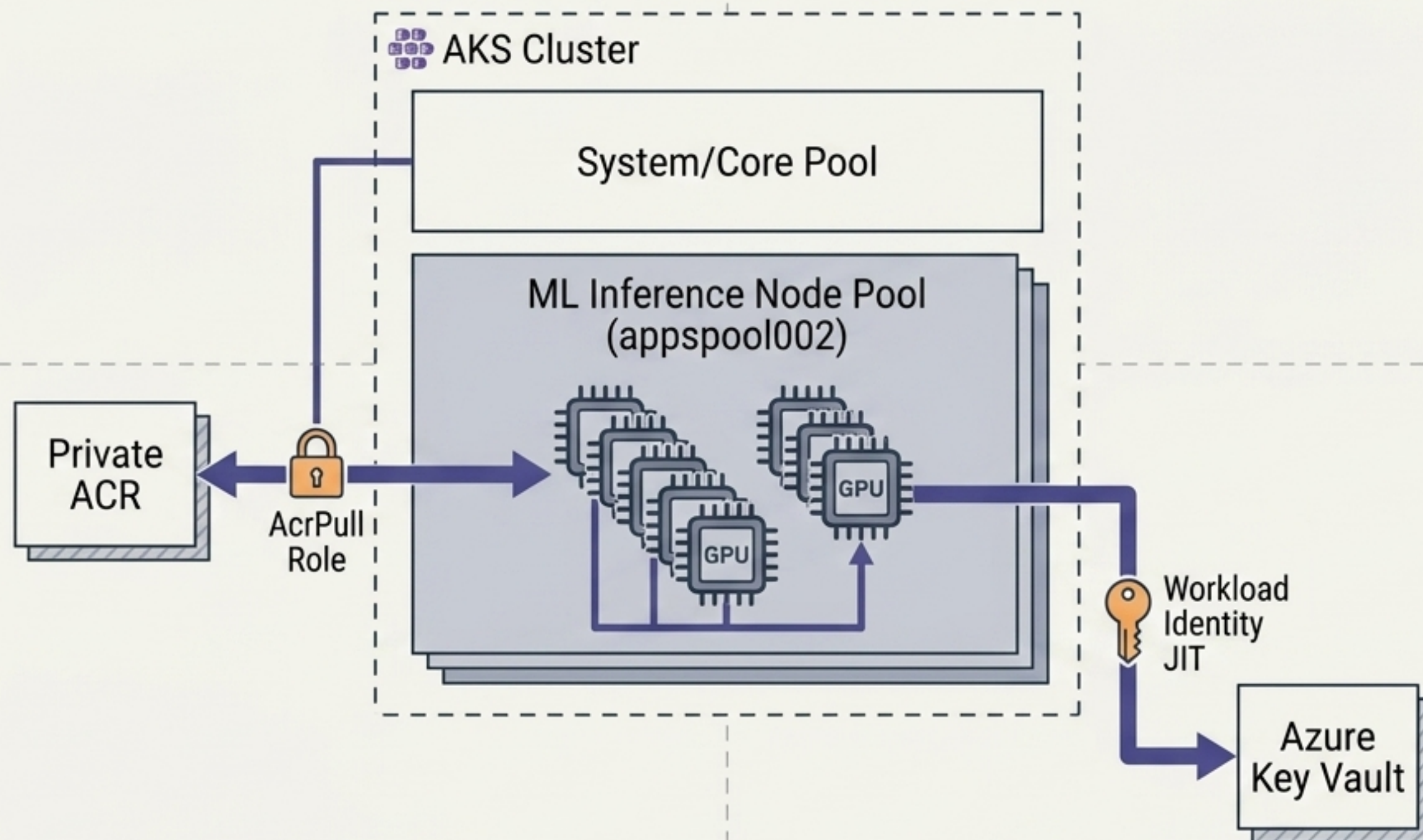
## Identity Flow

- User identity is maintained via MSAL on the frontend, transitioning to OAuth 2.0 OBO (On-Behalf-Of) flow to propagate context downstream.



# AKS: ML Inference Node Pool & Secure Access

Viewport (70%)





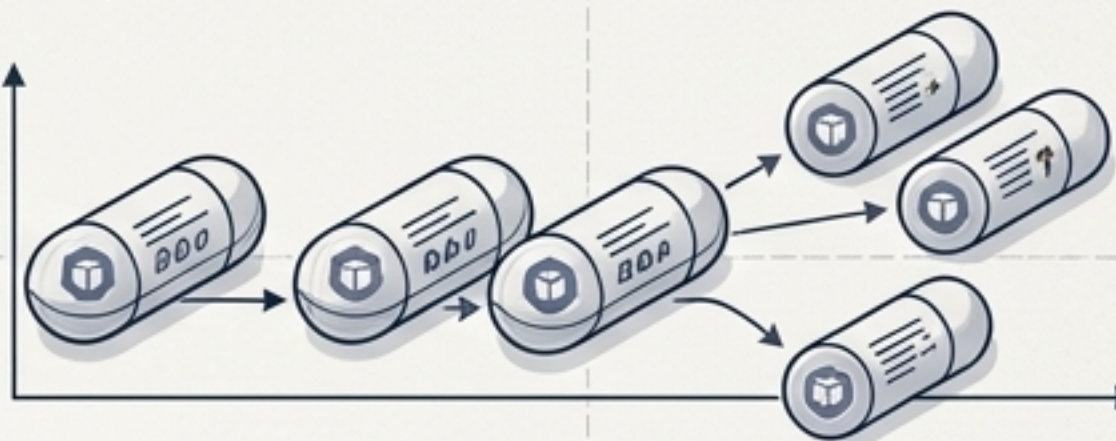
## Nodes (Compute Backbone)



**Component:** AKS Cluster Autoscaler (CA).

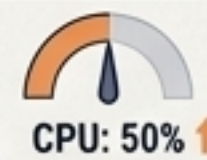
**Logic:** Triggers on pending pods. Prioritizes nodes handling the largest pending payload.

## Pods (Workload)



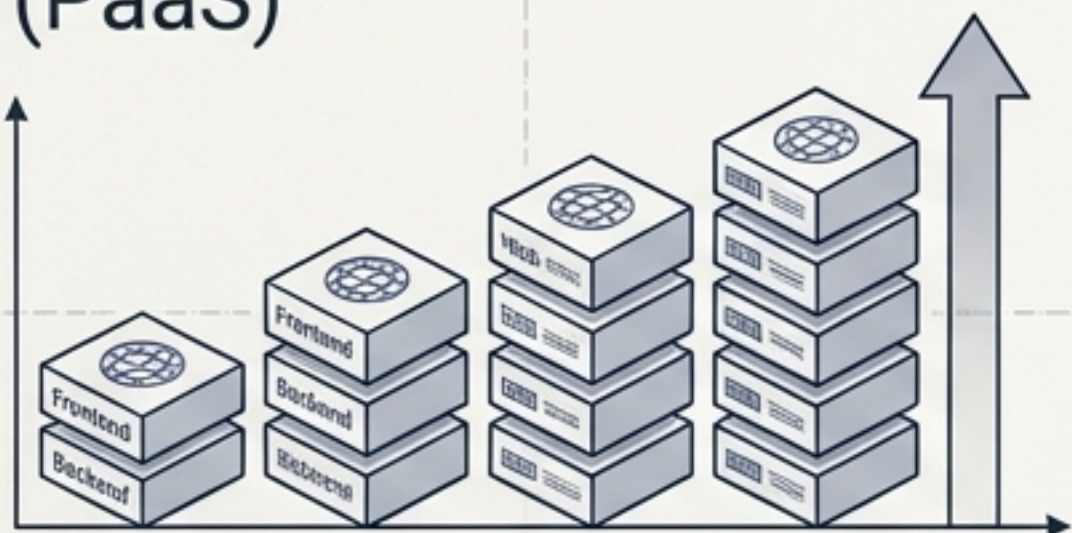
**Component:** AKS HPA / VPA.

**Logic:** Horizontal Pod Autoscaler targets 50% CPU utilization. Vertical Pod Autoscaler dynamically sizes based on telemetry.



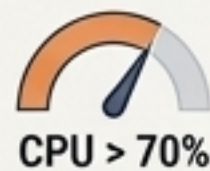
## 4-Tier Scaling Intelligence

## Web (PaaS)



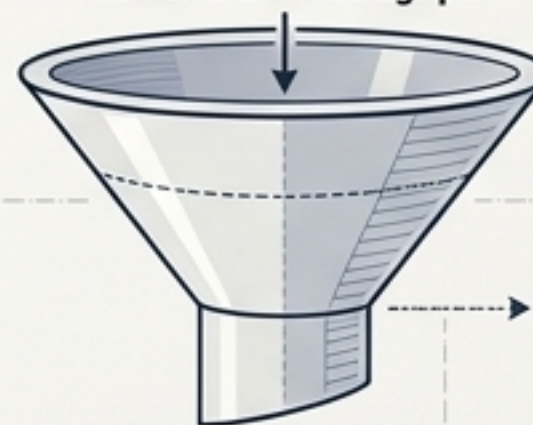
**Component:** App Service Elasticity.

**Logic:** Scale-out triggered automatically when CPU > 70% or RAM > 80%.



## Edge (WAF)

Inbound L7 Throughput



**Component:** Application Gateway.

**Logic:** Autoscale Units dynamically adjust based on inbound L7 throughput.



# Architectural Dependency & Provisioning Flow

Viewport (30%)

PROVISIONING SEQUENCE

## 1. Shared-Infra

Foundational Hub-Spoke,  
Core DNS, Azure Firewall.

## 2. AKS

Managed Clusters, Node  
Pools, OIDC setup.

## 3. App-Users

Entra ID user lifecycle  
and RBAC.

## 4. App-Core

Microservices, App  
Gateways, Persistence.

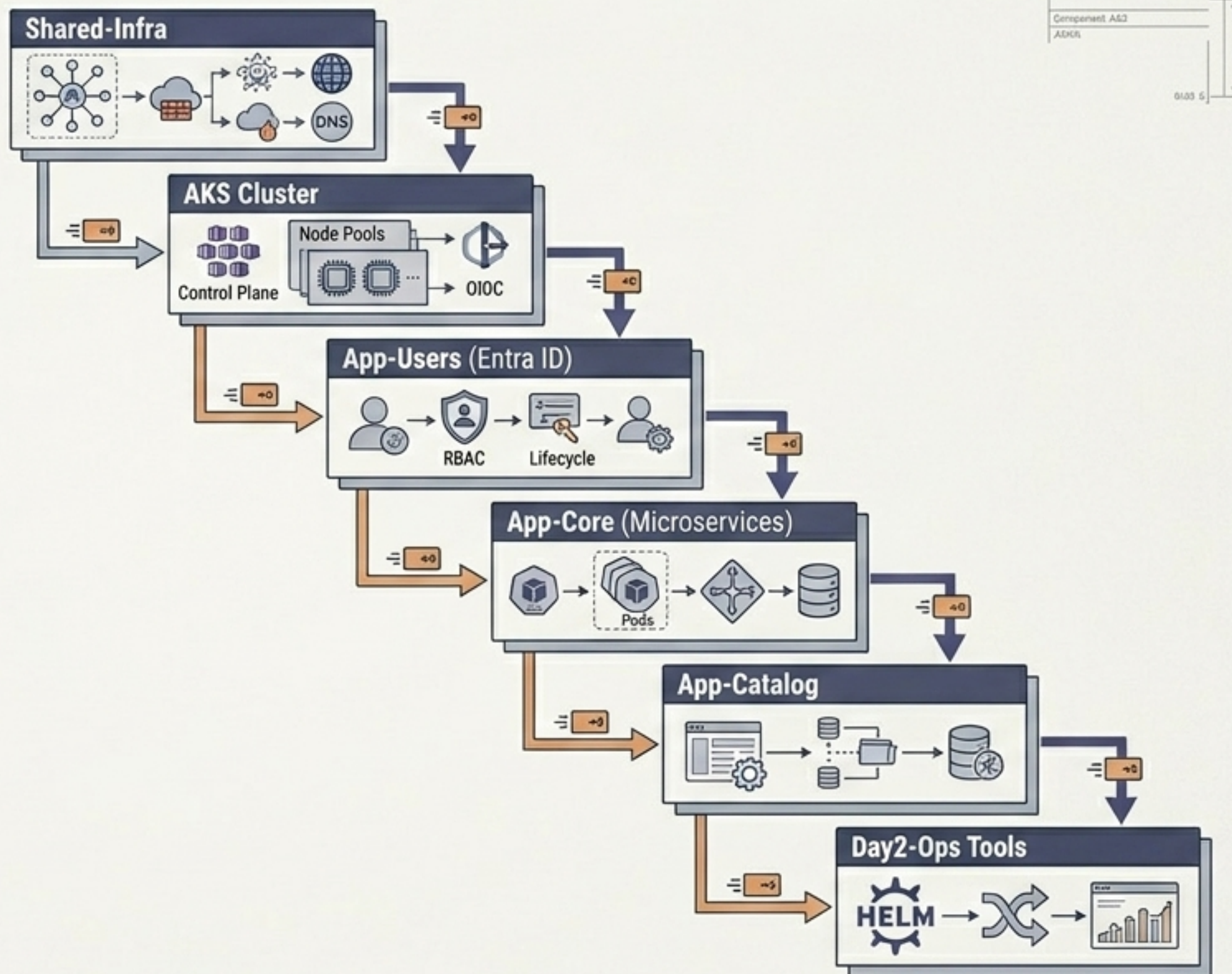
## 5. App-Catalog

Dynamic catalog and  
metadata services.

## 6. Day2-Ops

Helm, Ingress, Observability.

Viewport (70%)





Viewport (30%)

# The Immutable Deployment Execution

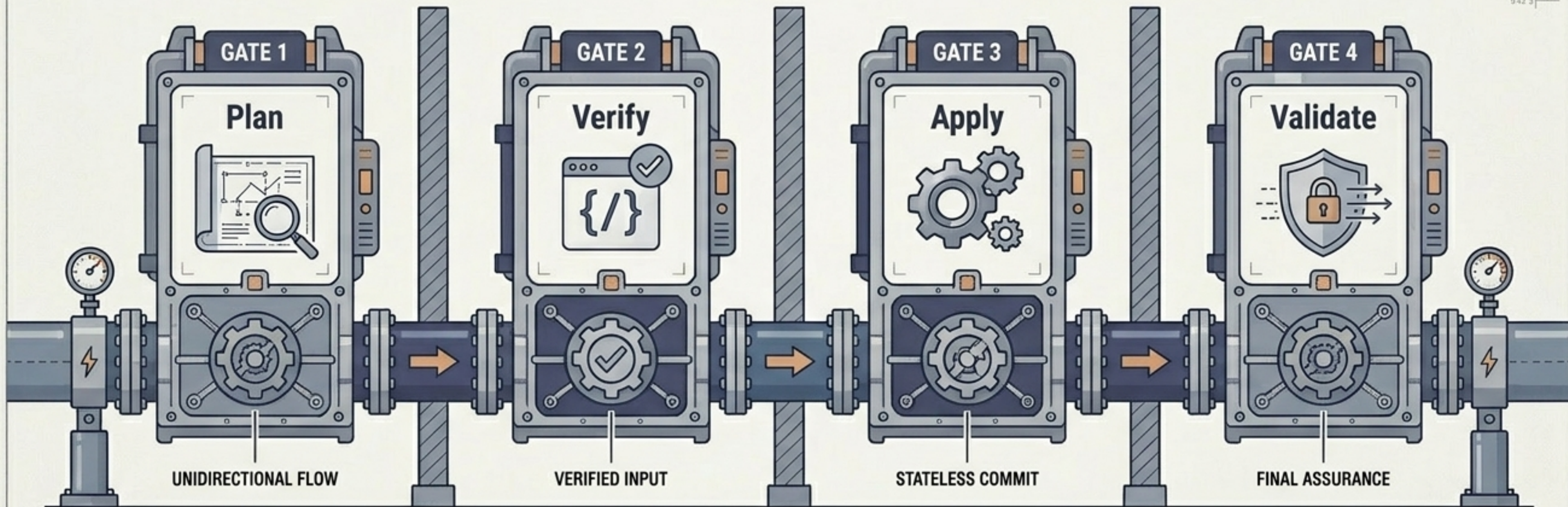
Strict **4-Stage Gate** mechanism ensuring a **Plan-First, Verify-Always** approach. Production is never compromised by unverified configuration drift, strictly treating infrastructure as stateless software.

Viewport (70%)

UNIDIRECTIONAL FLOW

NO BYPASS

FINAL ASSURANCE



STATEMENTS: 4-STAGE LOCKED EXECUTION

NO BYPASS

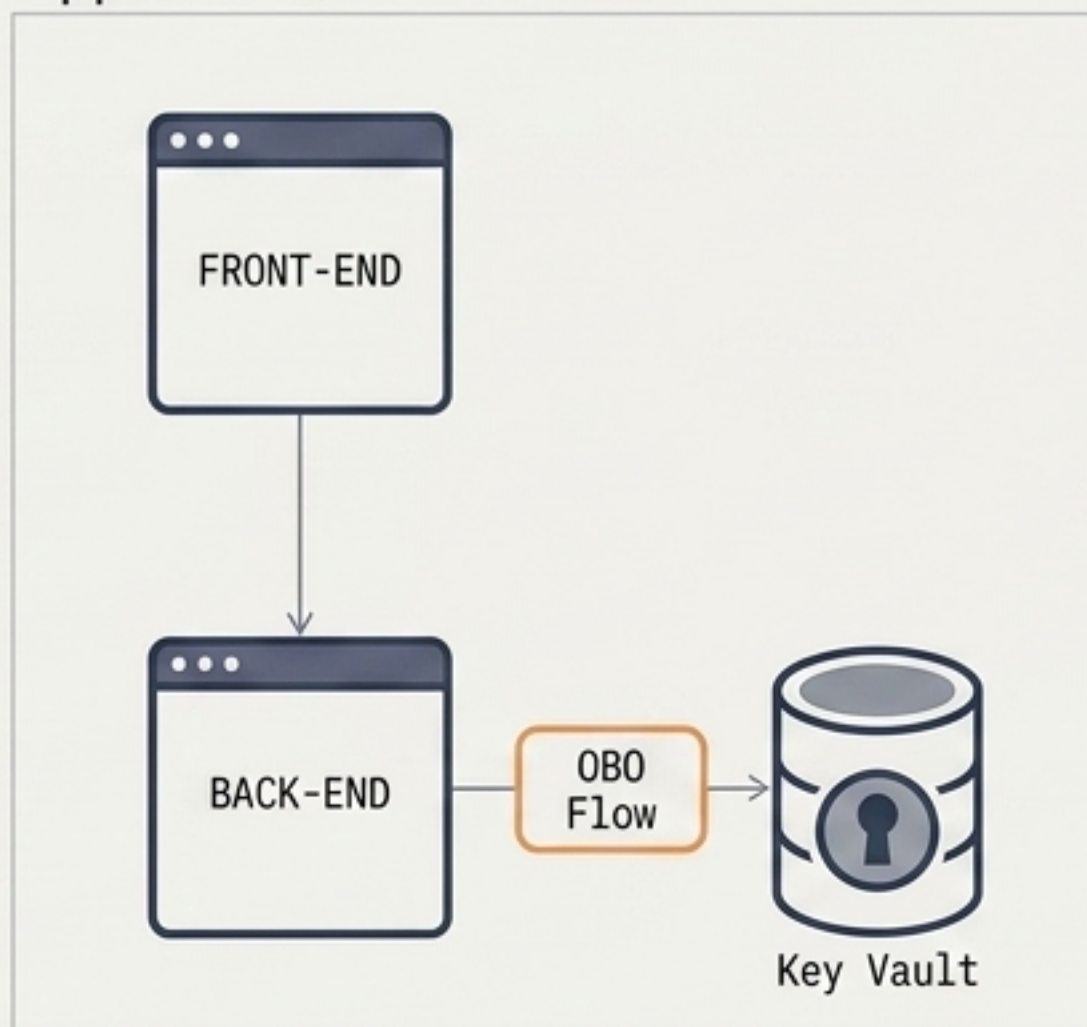
DEPLOYMENT INTEGRITY: MAX



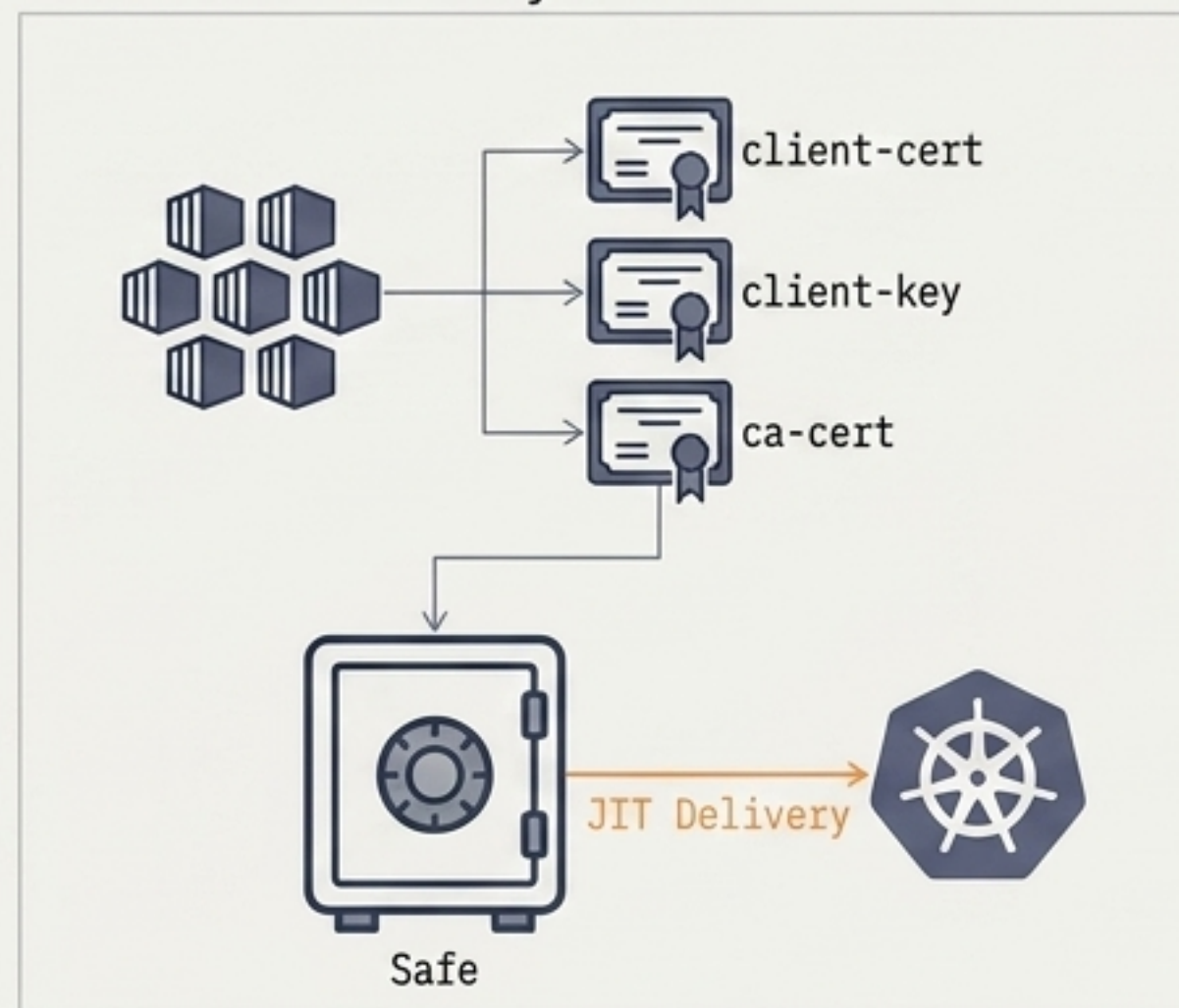
## Pipeline Secrets



## Application Secrets



## AKS Triad Lifecycle



## Access Protocol Matrix

|   | 1                   | 2                  | 3                                    |
|---|---------------------|--------------------|--------------------------------------|
|   | Secret Category     | Storage Location   | Access Protocol                      |
| 1 | Pipeline Secrets    | kv-sharedinfra-pro | Workload Identity (OIDC)             |
| 2 | Application Secrets | kv-{client}-{env}  | On-Behalf-Of (OBO) Flow              |
| 3 | AKS Credentials     | kv-sharedinfra-pro | Azure AD RBAC (Triad JIT extraction) |



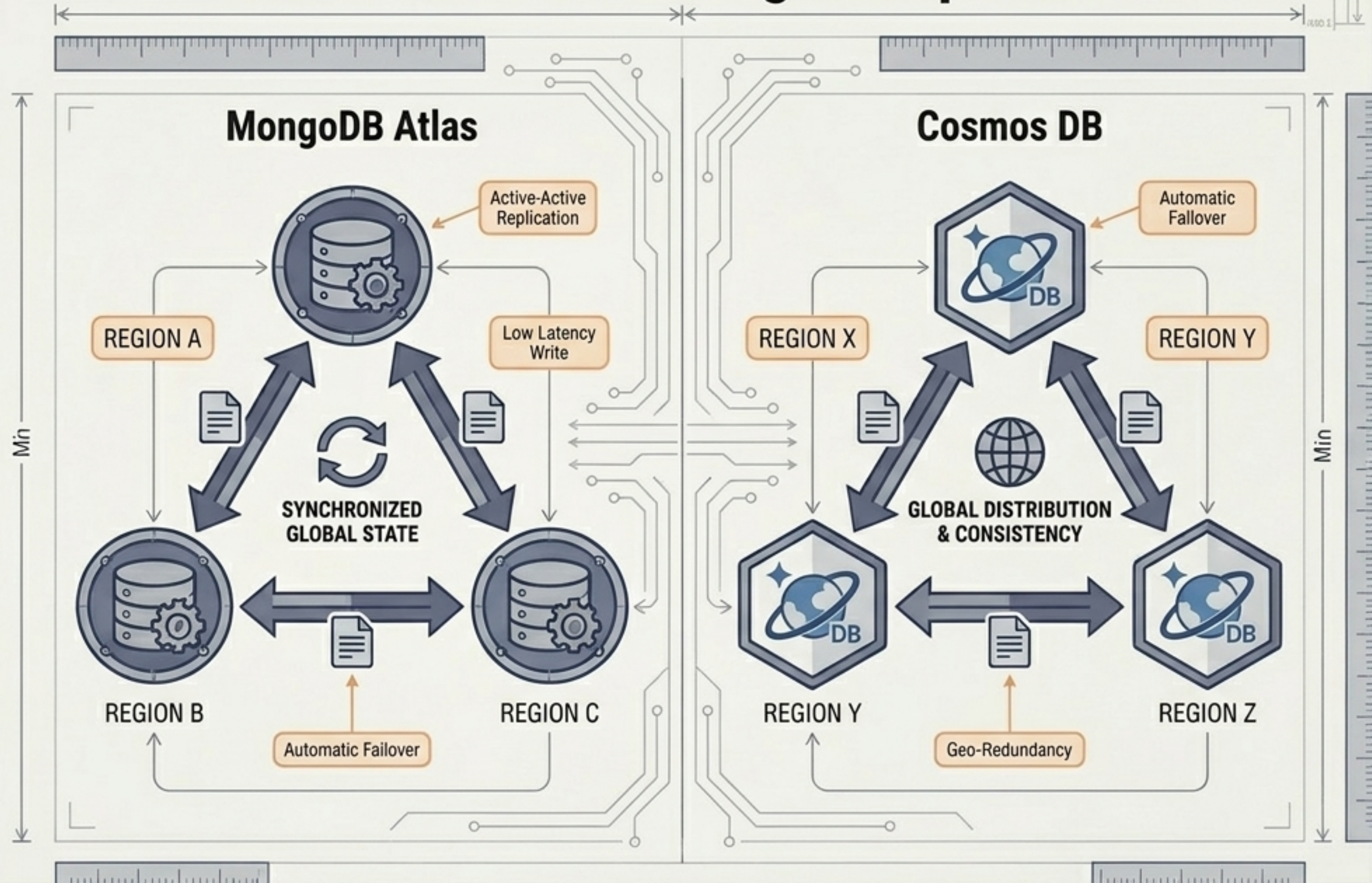
## Business Continuity Plan (BCP)

The persistence layer utilizes a **multi-cloud** and **multi-region** replication strategy to ensure uninterrupted data survival.

Automated vaulting and regional redundancy guarantee business continuity across geographical boundaries during **catastrophic failover scenarios**.

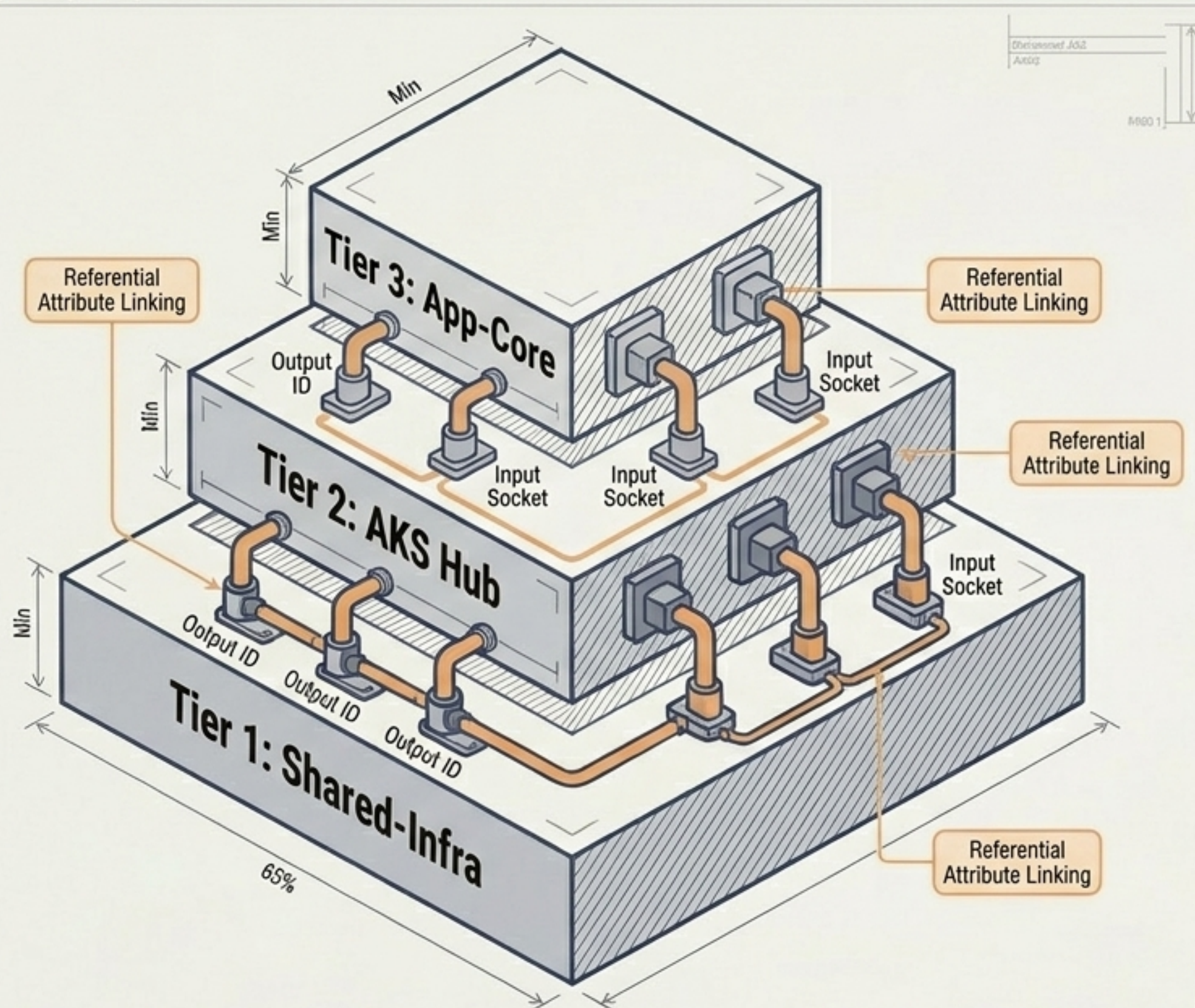
Viewport (75%)

# Multi-Cloud & Multi-Region Replication





Viewport (65%)



Explanation Panel (35%)

## Explanation Panel

### Composite over Atomic

Large, composite modules map directly to business domains rather than managing atomic, disparate resources.

### 3-Tier Logical Hierarchy

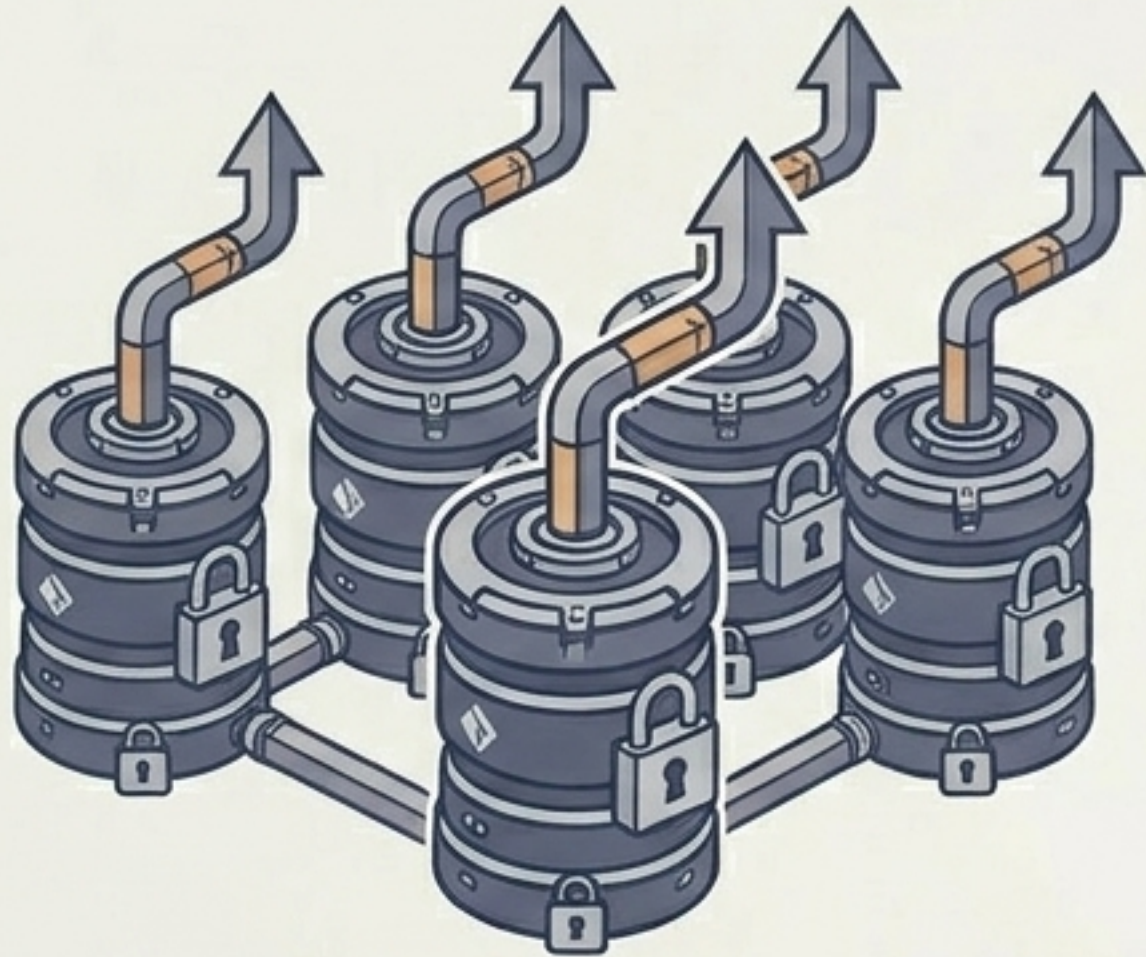
Decoupling into discrete tiers allows for partial lifecycle management without exposing the entire stack to blast-radius risk.

### Referential Attribute Linking

Code sovereignty achieved by passing attributes natively via HCL outputs rather than hardcoding resource names.

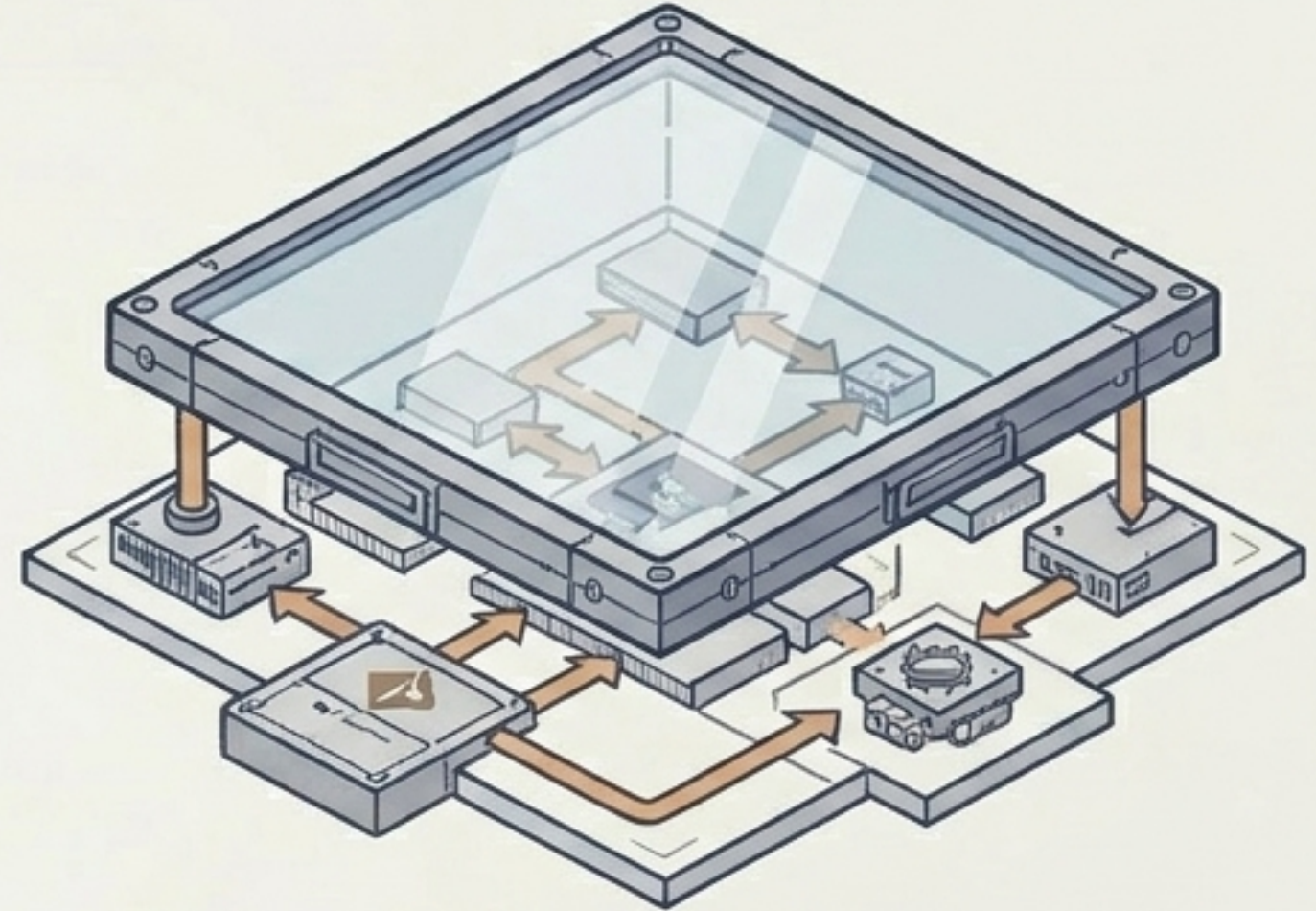


## Federated Multi-Repo (Production Execution)



- Independent state locks
- Minimized blast radius
- Fast CI/CD execution per isolated stack

## Mono-Repo (GitHub/Distribution)



- Acts as a Single Pane of Glass
- Total architectural visibility
- Comprehensive contextual understanding

**Conclusion:** Multi-Repo enforces execution speed and isolation; Mono-Repo provides distribution and system visibility.



# Terraform 1.15 Catalyst and Vision

Pipeline Orchestrator

## Current Implementation (Pipeline-Driven)



Terraform Process 1



Terraform Process 2



Terraform Process 3

Sequential, Disjointed Orchestration

## Future Vision (Native Terraform 1.15 Stacks)



AKS Hub

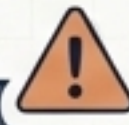


App-Core



Native Terraform  
Core Block

Formal Deprecation  
Warning



Dynamic  
Variables



Shared-Infra



Data Services

Unified, Native HCL Orchestration

## Explanation Panel

### The Catalyst (Terraform 1.15)

- **Dynamic Module Sources:** Variables in module blocks enable multi-region scaling without duplicated code.
- **Formal Deprecations:** Safe, warned module upgrades across the enterprise.
- **Type Constraints:** Strict variable contracts ensuring reliable data flow.

### Synthesis

Terraform 1.15 closes the gap, allowing the intricate dependencies of the Vision 2026 blueprint to be orchestrated entirely within native HCL, achieving true IaC sovereignty.